

Mathematical Analysis Lecture 20. 积分学:

e.g. $\int \frac{e^{3x}+1}{e^x+1} dx = \int \frac{(e^x+1)(e^{2x}+1-e^x)}{e^x+1} dx = \int (e^{2x}-e^x+1) dx = \frac{1}{3}e^{3x}-e^x+x+C, C \in \mathbb{R}$. 善用代数变换.

未来我们可以用换元积分法解决问题

换元积分法: 用于简化积分运算

设 $F'(u)=f(u)$, $u=\varphi(x)$ 可导, 则有 $dF(\varphi(x))=f(\varphi(x)) \cdot \varphi'(x) \cdot dx$

$\therefore \int f(\varphi(x)) \cdot \varphi'(x) dx = \int f(u) du = F(u) + C = F(\varphi(x)) + C, C \in \mathbb{R}$, 称为换元积分法.

$\int f(\varphi(x)) \cdot \varphi'(x) dx \xrightarrow[\text{第二类换元法}]{\text{第一类换元法}} \int f(u) du$

→ 第一类换元法: 关键在建换 $\varphi(x)$, 使 $\varphi'(x) dx = d\varphi(x)$

线性换元. e.g. $\int (ax+b)^m dx = \int (ax+b)^m \cdot \frac{1}{a} \cdot a \cdot dx = \int (ax+b)^m \cdot \frac{1}{a} (ax+b)' \cdot dx = \frac{1}{a} \int (ax+b)^m \cdot d(ax+b)$
 $= \frac{1}{a(m+1)} (ax+b)^{m+1} + C \quad (m \neq -1, m = -1 \text{ 时也可用换元法处理})$

e.g. $\int \frac{dx}{a^2+x^2} = \frac{1}{a^2} \int \frac{1}{1+(\frac{x}{a})^2} dx = \frac{1}{a^2} \int a \cdot \frac{1}{1+(\frac{x}{a})^2} d(\frac{x}{a}) = \frac{1}{a} \cdot \arctan(\frac{x}{a}) + C$. 联想基本积分表.

e.g. $\int \frac{dx}{\sqrt{a^2-x^2}} = \frac{1}{a} \int a \cdot \frac{1}{\sqrt{1-(\frac{x}{a})^2}} d(\frac{x}{a}) = \arcsin(\frac{x}{a}) + C$.

三角换元 e.g. $\int \tan x dx = \int \frac{\sin x}{\cos x} dx = - \int \frac{1}{\cos x} d\cos x = -\ln|\cos x| + C$.

e.g. $\int \frac{dx}{x^2-a^2} = \int \frac{dx}{(x+a)(x-a)} = \frac{1}{2a} \int (\frac{1}{x-a} - \frac{1}{x+a}) dx = \frac{1}{2a} \cdot \ln|\frac{x-a}{x+a}| + C$.

常见的换元形式: $\int f(ax+b) \cdot dx = \int f(ax+b) \cdot \frac{1}{a} d(ax+b)$

$\int f(x^n) x^{n-1} dx = \int f(x^n) \cdot \frac{1}{n} d(x^n)$

$\int f(\ln x) \cdot \frac{1}{x} dx = \int f(\ln x) d\ln x$

$\int f(x^n) \cdot \frac{1}{x} dx = \frac{1}{n} \int f(x^n) \frac{1}{x^n} \cdot d(x^n)$

$\int f(e^x) e^x dx = \int f(e^x) de^x$

$\int f(\cos x) \cdot \sin x = - \int f(\cos x) d\cos x$

$\int f(\tan x) \sec^2 x = \int f(\tan x) \cdot d(\tan x)$

e.g. $\int \frac{dx}{x(1+2\ln x)} = \frac{1}{2} \int \frac{1}{1+2\ln x} d(1+2\ln x) = \frac{1}{2} \ln|1+2\ln x| + C$

e.g. $\int \sec^6 x dx = \int (\tan^2 x + 1)^2 d\tan x = \frac{1}{5} \tan^5 x + \frac{2}{3} \tan^3 x + \tan x + C$

e.g. $\int \frac{dx}{e^x+1} = \int \frac{e^x+1-e^x}{e^x+1} = \int dx - \int \frac{d(1+e^x)}{1+e^x} = x - \ln(1+e^x) + C$
 $= \int \frac{e^{-x}}{1+e^{-x}} = -\ln(1+e^{-x}) + C$

e.g. $\int \sec x dx = \int \frac{1}{\cos x} dx = \int \frac{1}{\cos^3 x} d\sin x = \int \frac{1}{1-\sin^2 x} d\sin x = \frac{1}{2} \ln|\frac{1+\sin x}{1-\sin x}| + C$
 $= \int \sec x \cdot \frac{\tan x + \sec x}{\tan x + \sec x} dx = \int \frac{d(\tan x + \sec x)}{\tan x + \sec x} = \ln|\tan x + \sec x| + C$

e.g. $\int \frac{x^3}{(x^2+a^2)^{3/2}} dx = \int \frac{x^2}{(x^2+a^2)^{3/2}} \cdot \frac{1}{2} dx^2 = \frac{1}{2} \int \frac{(x^2+a^2)-a^2}{(x^2+a^2)^{3/2}} dx^2 = \dots$

e.g. $\int \cos^4 x dx = \int (\frac{1}{4}(\frac{3}{2} + 2\cos 2x + \frac{1}{2}\cos 4x)) dx = \dots$

e.g. $\int \frac{dx}{x^2+2x+3} = \int \frac{dx}{(x+1)^2+2} = \dots$

e.g. $\int \frac{x+1}{x(1+xe^x)} dx = \int \frac{(x+1)e^x}{xe^x(1+xe^x)} dx = \int \frac{1}{xe^x(1+xe^x)} d(xe^x) = \ln|\frac{xe^x}{1+xe^x}| + C$

e.g. $\int \frac{dx}{x(x^{10}+1)} = \int \frac{x^9}{x^{10}(x^{10}+1)} dx = \frac{1}{10} \int \frac{1}{x^{10}(x^{10}+1)} d(x^{10}) = \dots$ (法1) 上面乘一个, 下面乘一个
 $= \int \frac{1+x^{10}-x^{10}}{x^{10}(x^{10}+1)} dx = \int \frac{1}{x} dx - \int \frac{x^9}{x^{10}+1} dx = \dots$ (法2) 加一个减一个
 $= \int \frac{dx}{x^{11}(x^{10}+1)} = -\frac{1}{10} \int \frac{1}{x^{10}+1} d(x^{10}) = \dots$